EXPERIMENT-7

7. Scenario: You are working as a data analyst for an e-commerce company. You have been given a dataset containing information about customer orders, stored in a Pandas DataFrame named order_data. The DataFrame has columns for customer ID, order date, product name, and order quantity. Your task is to analyze the data and answer specific questions about the orders.

Question: Using Pandas DataFrame operations, how would you find the following information

from the order_data DataFrame:

1. The total number of orders made by each customer.

2. The average order quantity for each product.

3. The earliest and latest order dates in the dataset.

Aim

To analyze customer order data using Pandas DataFrame operations and find the total orders made by each customer, average order quantity for each product, and earliest and latest order dates.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing customer IDs, order dates, product names, and order quantities.
3. Use groupby() and count() to find the total orders made by each customer.
4. Use groupby() and mean() to calculate the average order quantity for each product.
5. Convert the order dates into datetime format.
6. Use min() and max() to find the earliest and latest order dates.
7. Display all the results.

Code

```
import pandas as pd

order_data = pd.DataFrame({
    'Customer_ID': [101, 102, 101, 103, 102, 101],
    'Order_Date': [
        '2026-09-01', '2026-09-03', '2026-09-05',
        '2026-09-02', '2026-09-07', '2026-09-10'
    ],
    'Product_Name': [
        'Laptop', 'Mouse', 'Mouse',
        'Keyboard', 'Laptop', 'Keyboard'
    ],
    'Order_Quantity': [1, 2, 3, 1, 2, 4]
})
order_data['Order_Date'] = pd.to_datetime(
    order_data['Order_Date']
)
# 1. Total orders made by each customer
customer_orders = order_data.groupby(
    'Customer_ID'
).size()
```

```
# 2. Average order quantity for each product
average_quantity = order_data.groupby(
    'Product_Name'
)['Order_Quantity'].mean()
# 3. Earliest and latest order dates
earliest_date = order_data['Order_Date'].min()
latest_date = order_data['Order_Date'].max()
print("Total orders by each customer:")
print(customer_orders)
print("\nAverage order quantity for each product:")
print(average_quantity)
print("\nEarliest order date:", earliest_date.date())
print("Latest order date:", latest_date.date())
```

Input

Customer_ID	Order_Date	Product_Name	Order_Quantity
101	2026-09-01	Laptop	1
102	2026-09-03	Mouse	2
101	2026-09-05	Mouse	3
103	2026-09-02	Keyboard	1
102	2026-09-07	Laptop	2

101 2026-09-10 Keyboard 4

Output

Total orders by each customer:

Customer_ID

101 3

102 2

103 1

Average order quantity for each product:

Product_Name

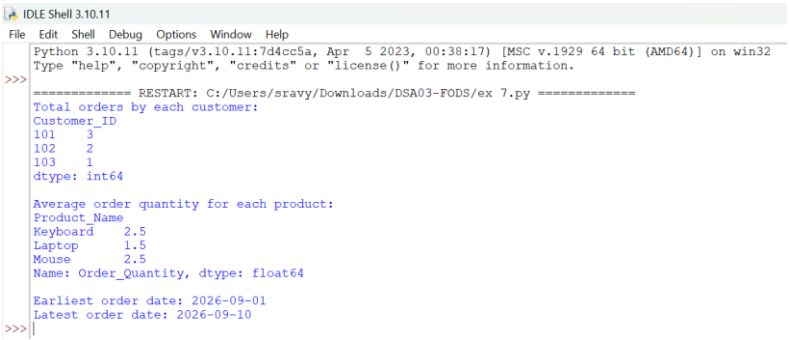
Keyboard 2.5

Laptop 1.5

Mouse 2.5

Earliest order date: 2026-09-01

Latest order date: 2026-09-10



Result

Thus, the total number of orders made by each customer, average order quantity for each product, and earliest and latest order dates were determined successfully using Pandas.